MILITARY STANDARD

PRINTED-WIRING AND PRINTED-CIRCUIT
TERMS AND DEFINITIONS

TO ALL HOLDERS OF MIL-STD-429C

1. THE FOLLOWING PAGES OF MIL-STD-429C HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
5	25 November 1974	(REPRINTED WITHOUT CHANGE)	
6, 6a		6	25 November 1974
21	25 November 1974	(REPRINTED WITHOUT CHANGE)	
22		22	

2. THE FOLLOWING PAGES HAVE BEEN ADDED:

NEW PAGES	DATE
23 through 37	

3. RETAIN THIS NOTICE AND INSERT BEFORE THE TABLE OF CONTENTS.

4. Holders of MIL-STD-429 will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the Military Standard is completely revised or canceled.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 11, 26, 99, 85
DLA - ES
NSA/S2

User activities:
Army - ME, MU, PA, WC
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(Project 5999-0083)

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3.45 Component mounting. Component mounting is the act of mechanically attaching the component to the printed board, or the manner in which they are attached, or both.

3.46 Component orientation. Component orientation is the direction in which the components, on a printed board or other assembly, are lined-up electrically with respect to the polarity of polarized components and also with respect to one another and to the board.

3.47 Component side. The component side is the side of a printed board upon which components are mounted.

3.48 Composite board. A composite board is a completely laminated multilayer printed board.

3.49 Conductive foil. See "Foil".

3.50 Conductor. A conductor is a single conductive line forming an electrical connection between terminal areas. (In printed board terminology, the terms "conductor", "conductor line", "conductor path", "line", "path", "trace", and "track" are synonymous; but the preferred term is "conductor".)

3.51 Conductor layer. A conductor layer is a layer of a printed board containing conductors. It also includes both ground planes and voltage planes.

3.52 Conductor layer no. 1. Conductor layer no. 1 is the conductor layer of a multilayer printed board on the component side.

3.53 Conductor pattern. A conductor pattern is a pattern of conductors in a conductor layer.

3.54 Conductor side. The conductor side is the side of a single-sided printed board containing the conductors.

3.55 Conductor spacing. Conductor spacing is the distance between adjacent edges of two adjacent conductors on any single conductor layer of a printed board. (See figure 5.)

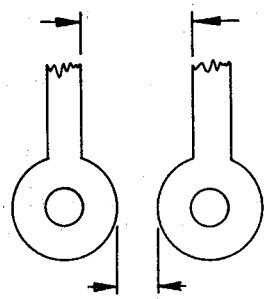


FIGURE 5. Conductor spacing.

3.56 Conductor thickness. Conductor thickness is the thickness of the conductor, exclusive of coatings or other metals.

3.57 Conductor width. Conductor width is the observable width of the conductor at any point chosen at random on the printed board viewed from vertically above. Imperfections, for example nicks, pinholes, or scratches, allowable by the relevant specification shall be ignored.

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3.58 Conformal coating. A conformal coating consists of a polymer film that conforms to the topography of the object coated and provides electrical insulation. It is used on printed-wiring assemblies to provide electrical insulation, resistance to moisture, water, salt water, fingerprints, oils, fuels, and solvents. To provide a minimum insulation resistance of 1,500,000 megohms, when tested in accordance with MIL-I-46058, the coating must be at least 0.0005 inch thick to provide the mechanical strength to resist cuts, scratches, abrasions, and cut-through abuse that would degrade the electrical insulation resistance.

3.59 Connection. A connection is the means by which electrical contact is made to the conductor pattern.

3.60 Connector keying pin. Connector keying pin is a pin that permits a plug to be plugged into its mating receptacle, but prevents it from being plugged into any other receptacle. (See also "Polarizing pin.")

3.61 Connector, printed-circuit, two-part. A two-part, printed-circuit connector consists of two plastic bodies containing the pin and socket electrical contacts, one of which mounts on the printed board (as a separate hardware item) and the other mounts on a chassis or other hardware and accepts the first for making electrical contact. There are two major categories, as follows:

a. Hook-up-wiring receptacle.
b. Printed-board plug or receptacle.

3.62 Connector receptacle, printed-board, one-part. A one-part, printed-board, connector receptacle is a connector receptacle containing contacts that accept the edge of the printed board upon which the printed wiring serves as the mating contacts (see "Printed contact"). A receptacle of this type is also called a "card-edge connector", "card-insertion connector", "one-part printed circuit connector", or "one-piece printed-circuit connector."

3.63 Control drawing. Use "Master drawing."

3.64 Coupon. A coupon is a specimen of a printed board, or of printed-board material, for testing purposes. (See also "Quality-conformance-test coupon".)

3.65 Cover layer. The cover layer is the outer layer or layers of insulating material over the conductors.

3.66 Crazing (base material). Crazing is an internal condition occurring in the laminated base material in the form of connected white spots or crosses (measling). It is usually made visible by mechanical or thermal stresses or both. This connected measling forms continuous paths which may carry moisture, or electric current, or both.

3.67 Crazing (conformal coating). Crazing is a network of fine cracks, extending on or under the surface of, or through a layer of the conformal coating.

3.68 Critical defect. A critical defect is a defect that judgment and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgment and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle. (Source: MIL-STD-105)

3.69 Cross hatching. Cross hatching is the breaking up of large conductive areas by use of a pattern of voids in the conductive material.

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3.70 Current-carrying capacity. Current-carrying capacity is the maximum current that can be continuously carried, under specified conditions by an individual conductor, without causing objectionable degradation of electrical or mechanical properties of the printed board.

3.71 Datum points. Datum points are points, assumed to be exact, for purposes of dimensional computation from which the location or geometric relationship of a printed-board feature may be established.

3.72 Defect. A defect is any nonconformance of the unit of product with specified requirements. (Source: MIL-STD-105)

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3.266 Unsupported hole. An unsupported hole is a hole containing no conductive material, plating, solder, or any type of reinforcement.

3.267 Unwetting. Use "Nonwetting."

3.268 Vapor deposition. Vapor deposition is a process consisting of condensation of a material from its gaseous state onto a base.

3.269 Via hole. A via hole is a plated-through hole which serves as a through connection between two or more internal conductor layers within a multilayer printed board.

3.270 Voltage plane. A voltage plane is a continuous sheet of metal on or in a printed board which functions at other than ground potential. It can also be used as a common point for circuit returns, shielding, and heat sinking.

3.271 Voltage-plane clearance. Voltage-plane clearance is the etched portion of a common voltage plane around a plated-through or nonplated-through hole which prevents connection between the voltage plane and the hole.

3.272 Warp (base material). Warp is the deviation from flatness of a board characterized by a roughly cylindrical or spherical curvature such that, if the board is rectangular, its corners or edges are in the same plane as the major surfaces of the board. (See figure 18.)

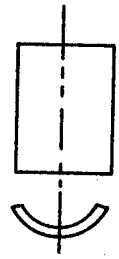


FIGURE 18. Warp (base material).

3.273 Warp (fabric). The warp consists of the threads of the reinforcing glass fabric that run in the machine or rolled direction of the basic roll of glass fabric when woven or processed. The resulting laminated base material is usually stronger in the warp direction than in the fill direction.

3.274 Wave soldering. Wave soldering is a process whereby printed boards are brought in contact with the surface of continuously flowing and circulating solder.

3.275 Weave exposure. Weave exposure is a surface condition in which the glass cloth is exposed (i.e. locally devoid of resin). It usually occurs because of chemical attack or mechanical abrasion. (See also "Fiber exposure.")

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3.276 Wetting. Wetting is the free flow and spreading of solder on a metallic surface to form an adherent bond. (See figure 19.)

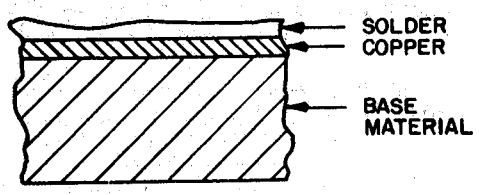


FIGURE 19. Wetting.

3.277 Whisker. A whisker is a slender acicular metallic growth which usually occurs after the printed board has been manufactured.

3.278 Wicking (chemical). Chemical wicking is the migration of chemicals into the glass fibers of the insulating material.

3.279 Wicking (solder). Solder wicking is the action of drawing up solder through capillary action.

3.280 Wiring pattern. Use "Conductor pattern."

3.281 Abut. To abut is to touch at one end, border (on, upon, or against).

3.282 Acceptable quality level. The acceptable quality level (AQL) is the maximum percent defective, for the purposes of sampling inspection, that can be considered satisfactory as a process average.

3.283 Acceptance inspection. Acceptance inspection consists of examination and testing to determine conformance of the supplies or services to certain specified requirements that serve as a basis for acceptance.

3.284 Adhesion. Adhesion is the state in which two surfaces are held together by interfacial forces consisting of chemical or mechanical forces, or both.

3.285 Adhesion promotion. Adhesion promotion is the chemical process of preparing a plastic surface to provide for a uniform, well-bonded metallic overplate.

3.286 Anchoring spurs. Anchoring spurs are hold down tabs on the terminal areas of flexible printed wiring, that extend beneath the cover layer and assist in holding the pads to the base. (See figure 20.)

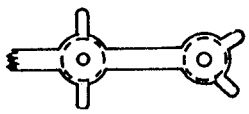


FIGURE 20. Pads with anchoring spurs.

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3.287 Axial leads. Axial leads are lead wires that are mounted along the axis or center line of a component or module.

3.288 Azeotrope. An azeotrope is a constant boiling point mixture of liquids.

3.289 Bellows contact. A bellows contact is a connector contact in which there is a flat spring folded to provide a uniform spring rate over the full tolerance range of the mating unit.

3.290 Bias. A bias is a line diagonal to the grain of a fabric.

3.291 Bifurcated contact. A bifurcated contact is a connector contact (usually a flat spring) that is slotted lengthwise to provide additional, independently-operating points of contact.

3.292 Birdcage. A birdcage is a defect in stranded wire where the strands in the stripped portion between the covering of an insulated wire and a soldered connection (or an end-tipped lead) have separated from the normal lay of the strands.

3.293 Bleeding. Bleeding is a condition in which a plated hole discharges process material or solution from crevices or voids.

3.294 Breakdown. Breakdown is a disruptive discharge through insulation involving a sudden and large increase in current through the insulation due to complete failure.

3.295 Bubble. A bubble is a small globule typically hollow and light: (1) a small body of gas within a liquid; (2) a thin film of liquid inflated with air or gas; (3) a globule in a transparent solid.

3.296 Bulge. A bulge is a swelling of a board usually caused by internal delamination or separation of fibers.

3.297 Bus bar. A bus bar is a conduit, such as a conductor on a printed board, to distribute electrical energy.

3.298 Camber. Camber is the planar deflection of a flat cable or flexible laminate from a straight line of specified length (a flat cable or flexible laminate with camber is similar to the curve of an unbanked race track).

3.299 Capacitive coupling. Capacitive coupling is the electrical interaction between two conductors caused by the capacitance between them.

3.300 Card. A card is a printed-wiring board. It is a completely processed conductor pattern or patterns, all formed on a common base.

3.301 Certification. Certification is the act of verifying that required training has been completed, and specified proficiency has been demonstrated.

3.302 Chamfer. Chamfer is a broken corner to eliminate an otherwise sharp edge.

3.303 Characteristic impedance. Characteristic impedance is the ratio of voltage to current in a propagating wave, i.e. the impedance that is offered to this wave at any point of the line. (In printed wiring its value depends on the width of the conductor, the distance from the conductor to ground plane(s), and the dielectric constant to the media between them).

3.304 Chemical stripping. Chemical stripping is the process of removing insulation from wire using chemical compounds.

3.305 Circuitry layer. A circuitry layer is a layer of a printed board containing conductors. It also includes both ground planes and voltage planes.

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3.306 Circuit continuity. Circuit continuity is the presence of a continuous path for the flow of electric current in a circuit.

3.307 Clinched wire interfacial connection. A clinched wire interfacial connection is a connection made by a wire that is passed through a hole in a printed circuit board and subsequently formed, or clinched against the conductor pattern on each side of the board and soldered.

3.308 Cold solder joint. A cold solder joint is a type of defective solder joint that occurs when the surface being soldered is not heated sufficiently and solder solidifies before wetting and flow can take place. The solder may appear smooth and shiny, but it has poor adhesion.

3.309 Component density. Component density is the quantity of components on a printed board per unit area.

3.310 Component hole. A component hole is a hole used for the attachment of an electrical connection of component terminations, including pins and wires, to the printed board.

3.311 Component lead. A component lead is the solid or stranded wire that extends from and serves as a connection to a component.

3.312 Conditioning. Conditioning is time-limited exposure of a test specimen to a specified environment(s) prior to testing.

3.313 Conduction soldering. Conduction soldering is a method of soldering that employs a soldering iron, or other contact methods, for transfer of heat to the soldering area.

3.314 Conductor base width. The conductor base width is the conductor width at the plane of the surface of the base material. (See figure 21.)

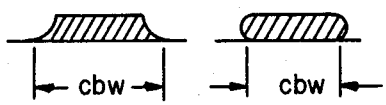


FIGURE 21. Conductor base width.

3.315 Conductor to hole spacing. Conductor to hole spacing is the distance between the edge of a conductor and the edge of a supported or unsupported hole.

3.316 Connector area. The connector area is that portion of printed wiring used for the purpose of providing external electrical connections.

3.317 Contact area. The contact area is the common area between a conductor and a connector through which the flow of electricity takes place.

3.318 Contact resistance. Contact resistance is the electrical resistance of the metallic surfaces at their interface in the contact area under specified conditions.

3.319 Contact spacing. Contact spacing is the distance between the center lines of adjacent contact areas.

3.320 Continuity. Continuity is the presence of a continuous path for the flow of electric current in a circuit.

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3.321 Contracting agency. The contracting agency is the organizational element of the Government that contracts for articles, supplies, or services, or it may be a contractor or subcontractor when the organizational element of the Government has given specific written authorization to such contractor or subcontractor to serve as agent for the contracting agency. A contractor or subcontractor serving as agent for the contracting agency shall not have the authority to grant waivers, deviations, or exceptions to the contract unless specific written authorization to do so has also been given by the Government organizational element.

3.322 Corner mark (crop mark). The corner mark is the mark at the corners of a printed board artwork, the inside edges of which usually locate the borders and establish the contour of the board.

3.323 Corrosion. Corrosion is the deterioration of metal by conversion to a chemical compound as a result of reacting with oxygen or other reactive materials, e.g. SO_2, Cl_2, etc., usually in the presence of moisture.

3.324 Cracking. Cracking is that phenomena manifest in coatings by a break extending through to the base surface.

3.325 Crosstalk. Crosstalk is the undesirable interference caused by the coupling of energy between signal paths.

3.326 Current-carrying capacity. Current-carrying capacity is the maximum current that can be continuously carried, under specified conditions, by an individual conductor without causing objectionable degradation of the electrical insulating characteristics or the mechanical properties of the printed boards.

3.327 Degradation. Degradation is deterioration, usually in the sense of a physical or chemical process rather than a mechanical one. There may be a specific amount of degradation permitted as a result of performance of environmental testing.

3.328 Desiccator. A desiccator is an air-tight container, containing a desiccant used for keeping parts or materials dry.

3.329 Design width of conductor. The design width of a conductor is the width of a conductor as delineated or noted on the master drawing.

3.330 Device. A device is a discrete electronic component consisting of one or more active or passive elements.

3.331 Dielectric constant. The dielectric constant of any homogeneous isotropic material is the ratio of the capacitance, C_X of a given configuration of electrodes with the material as the dielectric, to the capacitance, C_V, of the same electrode configuration with a vacuum (or air for most practical purposes) as the dielectric.

3.332 Dielectric withstanding voltage tests. Dielectric withstanding voltage tests are tests made to determine the ability of insulating materials and spacings to withstand specified overvoltages for a specified time without flashover or puncture.

3.333 Digitizing. Digitizing is any method of reducing feature locations on a flat plane to digital representation of X-Y coordinates.

3.334 Dimensioned hole. A hole in a printed board where the means of determining location is by coordinate values not necessarily coinciding with the stated grid.

3.335 Dissipation factor. The dissipation factor of an insulating material is the cotangent of the dielectric phase angle or the tangent of the dielectric loss angle. Since the power factor is the sine of the dielectric loss angle, the dissipation factor and the power factor are numerically the same for small loss angles.

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3.336 Disturbed solder connection. A disturbed solder connection is an unsatisfactory connection resulting from relative motion between the wire and termination point during solidification of the solder.

3.337 Drilling. Drilling is the cutting of holes, with a rotating twist drill that has two or more cutting edges meeting in the center of the cutting end and mounted concentric to the spindle.

3.338 Dual in-line package (DIP). A dual in-line package is a component that terminates in two straight rows of pins or lead wires.

3.339 E glass. E glass is a low alkali lime alumina borosilicate glass, noted for its good electrical properties. It has the following chemical formulation in percent by weight:

B_2O_3	5-10%	Na_2O and K_2O	0-2%
CaO	16-25%	TiO_2	0.-0.8%
Al_2O_3	12-16%	Fe_2O_3	0.05-0.4%
SiO_2	52-56%	F_2	0-1.0%

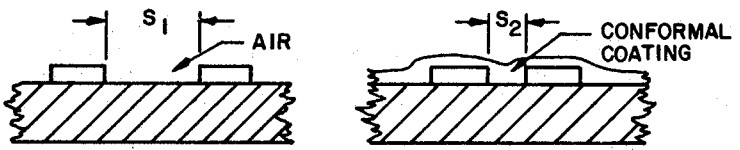
3.340 Edge definition. Edge definition is the fidelity of reproduction of a pattern edge relative to the production master.

3.341 Edge-dip test. The edge-dip test is a type of solderability test wherein a vertical specimen is immersed in molten solder, removed, and inspected.

3.342 Edge roughness. Edge roughness is roughness of the edges of printed-wiring conductors.

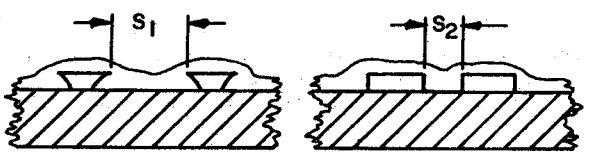
3.343 Electrical spacing. Electrical spacing is the allowable distance between adjacent conductors that is sufficient to prevent dielectric breakdown or corona, or both, between the conductors at any given voltage and altitude. The electrical spacing is dependent on the following factors (See figure 22.):

a. The dielectric strength of the air or the encapsulating material (e.g. conformal coating).



EFFECT OF DIELECTRIC STRENGTH ON ELECTRICAL SPACING, S.
$S_2 < S_1$ BECAUSE OF BETTER ELECTRICAL STRENGTH OF CONFORMAL COATING.

b. The edge configuration of conductors.



EFFECT OF EDGE CONFIGURATION ON ELECTRICAL SPACING, S.

FIGURE 22. Electrical spacing.

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3.344 Element (of a microcircuit or integrated circuit). A constitutent of the microcircuit or integrated circuit that contributes directly to its operation. (A discrete part incorporated into a microcircuit becomes an element of the microcircuit.)

3.345 End-item requirement. End-item requirements are those that specify the ultimate form of an item of production, but not the process by which it is manufactured.

3.346 Eutectic alloy. An eutectic alloy is a mixture of two or more metals that exhibit a single melting point and do not pass through a plastic state.

3.347 Extraneous copper. Extraneous copper is unwanted copper remaining on the base after chemical processing.

3.348 Eyelet. An eyelet is a short metallic tube, the ends of which can be bent outward and over to fasten it in place. (See figure 23.)

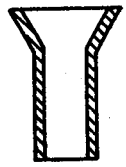


FIGURE 23. Eyelet.

3.349 Film integrated circuit. An integrated circuit consisting of elements that are films formed in situ upon an insulating substrate.

3.350 Fixture. A fixture is a device for supporting work during machining without guiding the cutting tools.

3.351 Flashover. Flashover is a disruptive discharge around or over the surface of an insulator, or between two charged surfaces not in contact with each other.

3.352 Fillet. A fillet is a blending or rounding of intersecting conductors or leads that eliminate sharp corners. Also, build-up of solder around a component lead, etc.

3.353 Flat cable. A flat cable is a cable with two or more parallel, round or flat, conductors in the same plane encapsulated by an insulating material.

3.354 Flexible printed circuit. Flexible printed wiring is a random arrangement of printed wiring and components utilizing flexible base materials with or without flexible cover layers.

3.355 Flexure failure. Flexure failure is a conductor failure due to repeated flexing that is indicated by an increase of resistance to a specified value for a specified time.

3.356 Foil-clad laminate. Foil-clad laminate is a dielectric sheet material clad on one or both sides with metal foil, used to make printed-wiring boards.

3.357 From-to list. From-to list is written wiring instructions in the form of a list indicating termination points.

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3.358 FSCM. FSCM is an abbreviation which means Federal Supply Code for manufacturers.

3.359 Funginert. Funginert means not supporting fungus growth.

3.360 Fused coating. Fused coating is a metallic coating (usually tin or solder alloy) that has been melted and solidified forming a metallurgical bond to the base metal.

3.361 Gel time. Gel time is the time expressed in seconds, required for the resin to change its physical state from a solid through a liquid to a solid state again due to the action of thermal input.

3.362 Glass-transition temperature. The glass-transition temperature is the center of the temperature range in which a noncrystalline solid changes from being glass-brittle to being viscous.

3.363 Hand soldering. Hand soldering is soldering, by hand, with a soldering iron.

3.364 Heat sink. A device for dissipating heat away from a component or assembly.

3.365 Heat sink tool. A device with jaws of conductive material, e.g., copper, or aluminum, that clamps on a lead to protect the component from the heat of soldering.

3.366 Heat sinking plane. A heat sinking plane is a continuous sheet of metal on or in a printed board that functions to dissipate heat away from heat sensitive components.

3.367 Hole density. Hole density is the quantity of holes in a printed board per unit area.

3.368 Hole location. Hole location is the dimensional location of the center of a hole.

3.369 Hole pattern. Hole pattern is the arrangement of all holes in a printed board.

3.370 Hot spot. The hot spot is the point of maximum temperature on the outside of a device or component.

3.371 Hybrid microcircuit. A microcircuit consisting of elements that are a combination of the film circuit type and the semiconductor types, or a combination of one or both of the types with discrete parts.

3.372 Immersion plating (galvanic displacement). Immersion plating is the chemical deposition of a thin metallic coating over certain base metals by a partial displacement of the base metal.

3.373 Inspection overlay. An inspection overlay is a positive or negative transparency made from the production master and used as an inspection aid.

3.374 Integrated circuit. A microcircuit consisting of interconnected elements inseparably associated and formed in situ on or within a single substrate to perform an electronic circuit function.

3.375 Ionizable contaminants. Ionizable contaminants are residues such as flux activators, body salt, etch residue that are capable of ionizing and increasing the conductivity of water.

3.376 Landless hole. A landless hole is a plated-through hole without a land(s).

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3.377 Layer registration. Layer registration is the relative position of one or more printed circuitry patterns or portions thereof, with respect to their desired locations on or within a multilayer printed wiring board.

3.378 Lead projection. Lead projection is the distance that a component lead protrudes through the printed board on the side opposite the mounted component.

3.379 Logic diagram. Logic diagram is a drawing that depicts the multistate device implementation of logic functions with logic symbols and supplementary notations, showing details of signal flow and control, but not necessarily the point-to-point wiring.

3.380 Lot. A lot is a collection of units of product bearing identification and treated as a unique entity from which a sample is to be drawn and inspected to determine conformance with the acceptability criteria.

3.381 Manufacturing date. The manufacturing date is the date on which all of the manufacturing operations have been completed and the unit of product is ready to be inspected.

3.382 Margin. The margin is the distance between the reference edge of a cable and the nearest edge of the first conductor.

3.383 Metal-clad laminate. Metal-clad laminate is a dielectric sheet material, clad on one or both sides with metal foil, used for making printed-wiring boards.

3.384 Metallization. Metallization is a deposited or plated thin metallic film used for its protective or electrical properties.

3.385 Microelectronics. Microelectronics is that area of electronic technology associated with or applied to the realization of electronic systems from extremely small electronic parts or elements.

3.386 Microcircuit. Microcircuit is a small circuit having a high equivalent circuit element density that is considered as a single part composed of interconnected elements on or within a single substrate to perform an electronic circuit function. (This excludes printed wiring boards, circuit card assemblies, and modules composed exclusively of discrete electronic parts.)

3.387 Microcircuit module. Microcircuit module is an assembly of microcircuits or an assembly of microcircuits and discrete parts, designed to perform one or more electronic circuit functions, and constructed so that, for the purposes of specification testing, commerce, and maintenance, it is considered indivisible.

3.388 Microsectioning. Microsectioning is the preparation of a specimen for the microscopic examination, usually by cutting out a cross-section of the material to be examined, and providing suitable preparation such as mounting, polishing, etching, etc.

3.389 Minimum annular ring. The minimum annular ring is the minimum amount of copper (at the narrowest point) between the edge of the hole, and the edge of the terminal area before plating of the hole.

3.390 Minimum average thickness. The minimum average thickness is the thinnest permissible thickness. It is based on the arithmetic average of several thickness measurements when the minimum individual thickness requirement is specified.

3.391 Misregistration. Misregistration is the lack of dimensional correllation between successively deposited patterns.

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3.392 Module. Module is a separable unit in a packaging scheme displaying regularity of dimensions.

3.393 Moisture resistance. Moisture resistance is the ability of a material to resist absorbing ambient moisture.

3.394 Monolithic integrated circuit. A monolithic integrated circuit is an integrated circuit consisting of elements formed in situ on or within a semiconductor substrate with at least one of the elements formed within the substrate.

3.395 Multichip microcircuit. Multichip microcircuit is a microcircuit consisting of elements formed on or within two or more semiconductor chips that are separately attached to a substrate.

3.396 Negative etchback. Negative etchback is a condition in which the exposed edges of interior copper conductors, in drilled holes, in a multilayer board, have been etched away until the inside diameter of the hole through the copper is larger than the inside diameter of the as-drilled hole through the board. (See figure 24.)

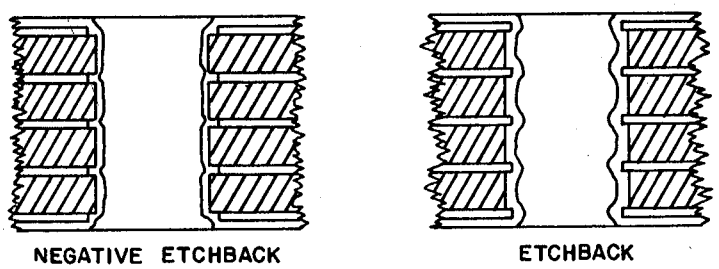


FIGURE 24. Negative etchback versus etchback.

3.397 Negative-working photoresist. Negative-working photoresist is a resist that is polymerized (hardened) by light and after exposure and development, remains on the surface of a laminate in those areas that were under the transparent parts of a production master.

3.398 Nick. A nick is a small cut in the edge of a conductor.

3.399 Nodule. A nodule is a rounded mass of irregular shape, a little lump.

3.400 Nominal cured thickness. The nominal cured thickness is the thickness of a laminate or multilayer board after the prepreg has been cured at the temperature and pressure specified for that particular class of resin flow.

3.401 Nonpolar solvents. Nonpolar solvents are solvents that will dissolve rosin, oil, wax, and grease.

3.402 Offset terminal area. Offset terminal areas are terminal areas on opposite sides of a printed-wiring board on opposite ends of the same hole that are offset so as not to border on the rim of the hole. (See figure 25.)

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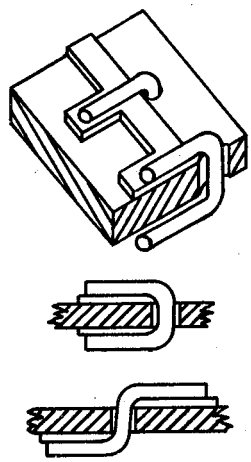


FIGURE 25. Offset terminal area.

3.403 Opaquer. An opaquer is a material that when added to the resin system, renders a laminate sufficiently opaque or translucent so that the yarn or weave of the reinforcing material cannot be seen with the unaided eye, using either reflected or transmitted light.

3.404 Outgassing. Outgassing is de-aeration or other gaseous emission from a printed board assembly (printed board, component, or connector) when exposed to a reduced pressure, or heat, or both.

3.405 Overplate. An overplate is the outer layer of plated metal.

3.406 Packaging density. Packaging density is the quantity of functions (components, interconnection devices, mechanical devices) per unit volume, usually expressed in qualitative terms, such as high, medium, or low.

3.407 Pattern. The pattern is the configuration of conductive and nonconductive materials on a panel or printed board. (Pattern denotes also the circuit configuration on related tools, drawings, and masters.)

3.408 Photographic reduction dimensions. Photographic reduction dimensions are dimensions (e.g., the distance between lines or between two specified points on the artwork master to indicate to the photographer the extent to which the artwork master is to be photographically reduced. (The value of the dimension refers to the 1:1 scale and must be specified.)

3.409 Pin density. The pin density is the quantity of pins on a printed board per unit area.

3.410 Pitch. Pitch is the nominal distance from center-to-center of adjacent flat cable conductors. (Where conductors are of equal size, and spacing is uniform, the pitch is usually measured from the reference edge of a conductor to the reference edge of the adjacent conductor.)

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3.411 Plate finish (pertaining to laminating). Plate finish is the finish present on the metallic surface of metal-clad base material resulting from direct contact with the laminating press plates without modification by any subsequent finishing process.

3.412 Plated-through-hole structure test. The plated-through-hole structure test is a visual examination of the metallic conductors and plated-through holes of a printed board after the glass-plastic laminate has been dissolved away.

3.413 Plating bar. A plating bar is a temporary conductive path interconnecting areas of a printed board to be electroplated, usually located on the panel outside of the borders of such a board.

3.414 Plating lot. A plating lot is defined as any number of boards or composite panels that are placed in any one plating tank and are processed through that one particular plating cycle.

3.415 Plotting. Plotting is the practice of mechanically converting X-Y positional information into a visual pattern, such as artwork.

3.416 Polarization. Polarization is a technique of eliminating symmetry within a plane so that parts can be engaged in only one way to minimize the possibility of electrical and mechanical damage, or malfunction.

3.417 Polarity. Polarity is a condition by which the direction of flow of current in a circuit can be determined.

3.418 Polar solvents. Polar solvents are solvents that are able to dissolve ionic salt residue.

3.419 Positive-working photoresist. A positive-working photoresist is a resist that is decomposed (softened) by light and, after exposure and development, is removed from those areas that were under the transparent parts of a production master.

3.420 Potting compound. A potting compound is an organic plastic used for encapsulation of components and/or wires.

3.421 Press-fit contact. A press-fit contact is an electrical contact that can be pressed into a hole in an insulator, printed board (with or without plated-through holes), or a metal plate.

3.422 Printed wiring assembly drawing. A printed wiring assembly document is a document that shows the printed board (rigid or flexible), the separately manufactured components that are to be added to the board, and any other information necessary to describe the joining of these parts to perform a specific function.

3.423 Printed wiring layout. A printed wiring layout is a sketch that depicts the printed wiring substrate, the physical size and location of electronic and mechanical components, and the routing of conductors that electrically interconnect components, in sufficient detail to allow the preparation of documentation and artwork.

3.424 Probe points. Probe points are predetermined locations on printed wiring boards where electrical contact may be made to exposed areas of the circuitry for electrical measuring purposes.

3.425 Production board. A finished board that the contractor is supplying in production.

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3.426 Production master. A one to one scale pattern that is used to produce one or more printed boards (rigid or flexible) within the accuracy specified on the master drawing.

3.427 Punching. Punching is the operation of removing a piece from a sheet stock by forcing a punch through the sheet and into a mating die.

3.428 Qualified products list (QPL). A qualified products list is a list of products, qualified under the requirements stated in the applicable specification, including appropriate product identification and test reference with the name and plant address of the manufacturer and distributer, as applicable.

3.429 Qualification. Qualification is the entire process by which products are obtained from manufacturers or distributors, examined and tested, and then identified on a qualified products list.

3.430 Quality conformance inspection. Quality conformance inspection is an inspection that includes all examinations and tests performed on items or services for the purpose of determining conformance with specified requirements.

3.431 Reference edge. The reference edge is the edge of a cable or conductor from which measurements are made. (Sometimes indicated by a thread, identification stripe, or printing. Conductors are usually identified by their sequential position from the reference edge, with number one conductor closest to this edge.)

3.432 Reflow melting. See "fusing".

3.433 Resin recession. Resin recession denotes the presence of voids between the barrel of the plated-through hole and the wall of the drilled hole, seen in micro-sections of plated-through holes in boards that have been exposed to high temperatures. (See figure 26.)

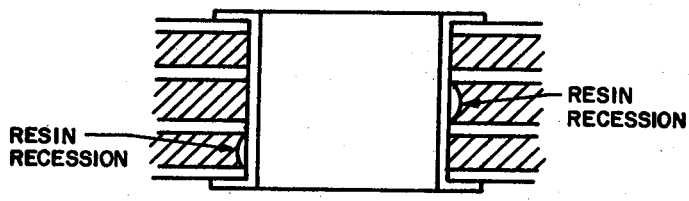


FIGURE 26. Resin recession.

3.434 Resin-rich. Resin rich denotes a significant thickness of nonreinforced surface-layer resin of the same composition as that within the base material.

3.435 Resin starvation. Resin starvation is a deficiency of resin in base material that is apparent after lamination by the presence of weave texture.

3.436 Resin starved area. A resin starved area is an area in a reinforced plastic article that has an insufficient amount of resin to wet out the reinforcement completely, evidenced by low gloss, dry spots, or exposed fibers.

3.437 Resistance soldering. Resistance soldering is a method of soldering in which a current is passed through and heats the soldering area by contact with one or more electrodes.

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3.438 Reverse etchback. Reverse etchback is a condition in which the exposed edges of interior copper conductors, in drilled holes, in a multilayer board, have been etched away until the inside diameter of the hole through the copper is larger than the inside diameter of the as-drilled hole through the board. (See figure 27.)

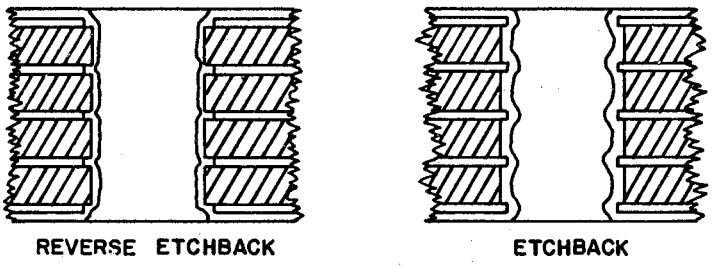


FIGURE 27. Reverse etchback versus etchback.

3.439 Reverse image. The reverse image is the resist pattern on a printed board used to allow for the exposure of conductive areas for subsequent plating.

3.440 Right-angle edge connector. A right-angle edge connector is a connector that terminates conductors at the edge of a printed board, while bringing the terminations out at right angles to the plane of the board conductors.

3.441 Roadmap. A roadmap is a printed pattern of nonconductive material by which the circuitry and components are delineated on a board to aid in service and repair of the board.

3.442 Robber. See "Thief".

3.443 Rosin solder connection. A rosin solder connection is an unsatisfactory connection having entrapped flux causing high electrical resistance.

3.444 Sampling plan. A sampling plan is a statement of the sample size or sizes to be used and the associated acceptance and rejection criteria.

3.445 Schematic diagram. A schematic diagram is a drawing that shows, by means of graphic symbols, the electrical connections, components, and functions of a specific circuit arrangement.

3.446 Scratch. A scratch is a relatively long, narrow, usually shallow furrow, or groove on a surface, caused by marking or rasping the surface with something pointed or sharp.

3.447 Shielding, electronic. Shielding, electronic is a physical barrier, usually electrically conductive, designed to reduce the interaction of electric or magnetic fields upon devices, circuits, or portions of circuits.

3.448 Signal. A signal is an electrical impulse of a predetermined voltage, current, polarity, and pulse width.

3.449 Signal conductor. A signal conductor is an individual conductor used to transmit an impressed signal.

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3.450 Signal plane. A signal plane is a conductor layer intended to carry signals, rather than serve as a ground or other fixed voltage.

3.451 Sliver. A sliver is a slender metallic projection that has been rent loose from the edge of a printed circuit conductor.

3.452 Solder. Solder is a single metal or an alloy of two or more metals that, when melted, is used to join metallic surfaces through the phenomena of wetting. Usually the major constituents are tin and lead.

3.453 Soldered connection. A soldered connection is an electrical/mechanical connection that employs solder for joining two or more metals.

3.454 Solder mask coating. A solder mask coating is a coating material used to mask or to protect selected areas of a pattern from the action of molten solder.

3.455 Solder plugs. Solder plugs are cores of solder in the plated-through holes of a printed-wiring board, used to improve the reliability of the board by improving the electrical continuity of the interfacial connections.

3.456 Solder side. Solder side is the side of a printed board that is opposite to the component side.

3.457 Solder spatter. Solder spatter is unwanted fragments of solder.

3.458 Solderability, edge dip test. Solderability, the edge dip, test is a type of solderability wherein a vertically mounted specimen is immersed in molten solder, removed, and inspected.

3.459 Soldering. Soldering is a process of joining metallic surfaces with solder, without the melting of the base metals.

3.460 Span. Span is the distance from the reference edge of the first conductor to the reference edge of the last conductor, expressed in decimal inches or centimeters.

3.461 Sparkover. Sparkover is a disruptive discharge between electrodes of a measuring gap.

3.462 Standoff terminal. A standoff terminal is a terminal generally postlike, having an axial portion of its body designed for projecting through or into a board for mounting. (See figure 28.)

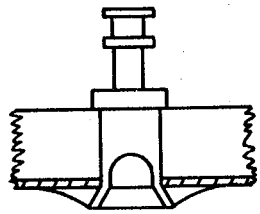


FIGURE 28. Standoff terminal.

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3.463 Straight-through lead. A straight-through lead is a component lead wire that is not clinched or swaged after insertion in a hole of a printed-wiring board. (See figure 29.)

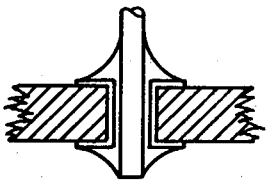


FIGURE 29. Straight through lead.

3.464 Step-and-repeat. Step-and-repeat is a method by which successive exposures of a single image are made to produce a multiple image production master.

3.465 Stress relief. Stress relief is a predetermined amount of slack to relieve tension in component or lead wires.

3.466 Strike. A strike is a thin film of metal to be followed by other coatings.

3.467 Stripline. A stripline is a type of transmission line configuration that consists of a single narrow conductor parallel and equidistant to two parallel ground planes.

3.468 Substrate (of a microcircuit or integrated circuit). Substrate is the supporting material upon or within which the elements of a microcircuit or integrated circuit are fabricated or attached.

3.469 Symbolization master. The symbolization master is an accurately scaled configuration showing the legend and symbols used to produce that sheet of the production master.

3.470 Tack time. Tack time is the point at which polymerization becomes visually apparent.

3.471 Taped components. Taped components are components attached to a continuous tape for and in automatic assembly.

3.472 Tenting. Tenting is a printed board fabrication method of covering over plated-through holes and the surrounding conductive pattern with a resist, usually dry film.

3.473 Terminal. A terminal is a tie-point device used to make electrical connections. Basic styles of terminals are: bifurcated, hook, perforated or pierced, solder cup, turret and straight post.

3.474 Terminal pull test. The terminal pull test is used to measure the bond strength of printed circuitry.

3.475 Test points. Test points are special points of access to an electrical circuit used for testing purposes. (See figure 30.)

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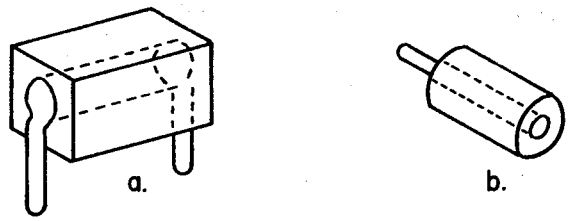


FIGURE 30. Printed wiring test points.

3.476 Thief. A thief is a racking device used in the electroplating process to provide a more uniform current density on plated parts. ("Thieves" absorb the unevenly distributed current on irregularly shaped parts, thereby assuring that the parts will receive an electroplated coating of uniform thickness.)

3.477 Thermal-shock test. The thermal-shock test is used to determine the ability of a printed-wiring board to withstand repeated temperature cycling without losing electrical continuity and current-carrying capacity, and without degradation of the materials.

3.478 Thermal-stress test. The thermal-stress test is used to determine the ability of a printed-wiring board to withstand the soldering operation without fracturing the plated-through holes or other conductors and without degradation of the materials. It consists of floating the specimens on molten solder followed by microsectioning the plated-through holes and visually examining the board and the microsectioned holes.

3.479 Thin laminate. A thin laminate is a laminated material that is thinner than 0.031 inches (0.079 cm).

3.480 Touching up. Touching up is the act of manually repeating a manufacturing operation for the purpose of improving the yield of acceptable parts.

3.481 Traceability. Traceability is the ability to trace a material or part through a manufacturers' operations and recognize and identify the material or part afterward.

3.482 Transmission cable. A transmission cable is two or more transmission lines. (If the structure is flat, it is called "flat transmission cable" to differentiate it from a round structure, such as a jacketed group of coaxial cables.) (See also Transmission line.)

3.483 Transmission line. A transmission line is a signal-carrying circuit composed of conductors and dielectric material with controlled electrical characteristics used for the transmission of high-frequency or narrow-pulse type signals.

3.484 Trim lines. Trim lines are lines that define the borders of a printed board. (See also "corner mark")

3.485 Visual examination. Visual examination is the qualitative observation of physical characteristics, using the unaided eye or stipulated levels of magnification.

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3.486 Void. A void is the absence of substance in a localized area.

3.487 Wrinkle. A wrinkle is a crease or fold in one or more outer layers of a laminated plastic (copperfoil, fabric, paper, conformal coating, etc.) that has been pressed in.

4. GENERAL REQUIREMENTS.

4.1 Not applicable.

5. DETAIL REQUIREMENTS.

5.1 Not applicable.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 11, 26, 99, 85
DLA - ES, GS
NS - S23

User activities:
Army - ME, MU, PA, WC
Navy - CG, OS
Air Force -

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5999-0060)

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